

A Novel Iterative Method for Principal Component Analysis

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Abstract—Principal Component Analysis (PCA) is a statistical method used for identifying key patterns from a given dataset and reducing its dimensions, while retaining most of the original information. The power method, which is one of the most well-known algorithms for PCA, is simple and easy to implement. However it obtains the specified number of principal components one by one. In this paper, we propose a novel iterative method for PCA, which is expressed as a system of nonlinear difference equations. Unlike the power method, the proposed method can obtain the specified number of principal components in parallel. We also show results of numerical experiments and theoretical analysis to verify the validity of the proposed method.

Index Terms—principal component analysis, power method, discrete-time nonlinear system, equilibrium point, stability.

I. INTRODUCTION

With the advancement of information and communication technology, it has been easier to acquire large amounts of data. Hence the importance of techniques for extracting key information from large-scale data is increasingly growing.

Principal Component Analysis (PCA) is one of the most fundamental data analysis methods for this purpose. Since Pearson first introduced it in his landmark paper [1], PCA has been developed into multiple variations tailored to diverse use cases [2]–[7], and has found various applications such as data compression [8], anomaly detection [9], climatology [10], data visualization [11], facial recognition [12], genetics [13].

The basic idea behind PCA is to identify the most meaningful subspace of a lower dimension such that the residual between the original data and their projections onto the subspace is minimized. Let $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_P\}$ be an orthonormal basis of the subspace. Then the projection of an original data sample $\mathbf{x} \in \mathbb{R}^N$ onto the subspace is expressed as $\sum_{i=1}^P y_i \mathbf{u}_i$. Namely, an N dimensional vector $\mathbf{x}$ can be transformed into a P dimensional vector $(y_1, y_2, \dots, y_P)^\top$.

The problem of finding an orthonormal basis of the most meaningful subspace is formulated as a constrained optimization problem. Moreover, solving this problem is equivalent to finding P normalized eigenvectors of the covariance matrix associated with the leading P eigenvalues. Therefore, the power method [14], a well-established numerical technique for finding dominant eigenvectors of symmetric matrices, has been used as a simple algorithm for PCA. The power method

first computes the first principal component by updating the value of a vector until it converges, and then subtracts it from the covariance matrix. The power method next computes the second principal component in the same way, and then subtracts it from the covariance matrix. This process is repeated until the P -th principal component is obtained.

Although the power method is simple and easy to implement, it has some disadvantages. Since the computation process of finding each principal component must be terminated after a finite number of iterations, the obtained value is different from the true value. Furthermore, these errors propagate to the computations of the subsequent principal components. To address this issue, this paper proposes a novel iterative method for PCA that can obtain the specified number of principal components in parallel. The proposed method is motivated by the continuous-time and discrete-time power methods [15], [16] for estimating the algebraic connectivity of a graph, and is described as a discrete-time nonlinear system. We first show through numerical experiments that the proposed method can find multiple principal components in parallel. We also perform theoretical analysis to find equilibrium points, determine their local stability, and show the boundedness of solutions.

II. PCA AND POWER METHOD

A. Principal Component Analysis

Let $\mathbf{X} = (\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_L) \in \mathbb{R}^{N \times L}$ be a data matrix consisting of L samples of dimension N . Throughout this paper, we assume that $N < L$ and the sum of all entries in each row of $\mathbf{X}$ is zero. Principal Component Analysis (PCA) is the process of identifying the most meaningful subspace of dimension $P (< N)$ such that the residual between the original data and their projections onto the subspace is minimized. Let $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_P\}$ be an orthonormal basis of an subspace of $\mathbb{R}^N$, and let $\mathbf{U} = (\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_P) \in \mathbb{R}^{N \times P}$ be the matrix of the basis. Then the projection of $\mathbf{x}_i$ onto the subspace is expressed as

$$\mathbf{U}\mathbf{U}^\top \mathbf{x}_i = \sum_{j=1}^P (\mathbf{x}_i^\top \mathbf{u}_j) \mathbf{u}_j,$$

which means that $\mathbf{x}_i$ is transformed into $\mathbf{y}_i = (y_{i1}, y_{i2}, \dots, y_{iP})^\top$ with $y_{ij} = \mathbf{x}_i^\top \mathbf{u}_j$ for $j = 1, 2, \dots, P$. Therefore, PCA

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is formulated as the following optimization problem:

$$\begin{aligned} & \text{minimize} \quad \|X - UU^\top X\|_F^2 \\ & \text{subject to} \quad U^\top U = I \end{aligned} \quad (1)$$

where $U = (\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_P)$ is the variable matrix, $\|\cdot\|_F$ denotes the Frobenius norm, and I is the identity matrix.

The covariance matrix of X is given by

$$S = \frac{1}{L} X X^\top \in \mathbb{R}^{N \times N}.$$

Since S is a symmetric matrix, there exist a diagonal matrix $\Lambda = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_N)$ and an orthogonal matrix $Q = (\mathbf{q}_1, \mathbf{q}_2, \dots, \mathbf{q}_N)$ such that

$$S = Q \Lambda Q^\top = \sum_{i=1}^N \lambda_i \mathbf{q}_i \mathbf{q}_i^\top \quad (2)$$

where $\lambda_1, \lambda_2, \dots, \lambda_N$ are the eigenvalues of S , and $\mathbf{q}_i$ is a normalized eigenvector of S associated with the eigenvalue λ_i . It is clear that S is a positive semi-definite matrix. So all the eigenvalues are nonnegative. In what follows, we assume for simplicity that

$$\lambda_1 > \lambda_2 > \dots > \lambda_P > \lambda_{P+1} \geq \dots \geq \lambda_N \geq 0. \quad (3)$$

Under this assumption, it is well known that $U = (\mathbf{q}_1, \mathbf{q}_2, \dots, \mathbf{q}_P)$ is an optimal solution of (1) [17]. Therefore, PCA reduces to finding P normalized eigenvectors of S associated with $\lambda_1, \lambda_2, \dots, \lambda_P$.

B. Power Method

Since an optimal solution to the optimization problem (1) can be obtained by extracting the leading P eigenvectors of the covariance matrix S , it is natural to consider using the power method [14], a well-established numerical technique for finding dominant eigenvectors of symmetric matrices.

The power method finds an approximate optimal solution of (1) in the following way. In the first step, a normalized vector $\mathbf{u}_1(0) \in \mathbb{R}^N$ is chosen and $\mathbf{u}_1(1), \mathbf{u}_1(2), \dots, \mathbf{u}_1(T_1)$ are obtained using the following rule:

$$\mathbf{u}_1(t+1) = \frac{S \mathbf{u}_1(t)}{\|S \mathbf{u}_1(t)\|}, \quad t = 0, 1, \dots, T_1 - 1.$$

The normalized vector $\mathbf{u}_1(T_1)$ is considered as an approximate eigenvector of S associated with λ_1 or an approximate first principal component of X . Thus $\mathbf{u}_1(T_1)$ is set to the first column of U , and the first principal component is subtracted from S as

$$\begin{aligned} S_2 &= S - S \mathbf{u}_1(T_1) \mathbf{u}_1(T_1)^\top \\ &= \sum_{i=1}^N \lambda_i \mathbf{q}_i \mathbf{q}_i^\top - \sum_{i=1}^N \lambda_i (\mathbf{q}_i^\top \mathbf{u}_1(T_1)) \mathbf{q}_i \mathbf{u}_1(T_1)^\top \\ &\approx \sum_{i=1}^N \lambda_i \mathbf{q}_i \mathbf{q}_i^\top - \lambda_1 \mathbf{q}_1 \mathbf{q}_1^\top \\ &= \sum_{i=2}^N \lambda_i \mathbf{q}_i \mathbf{q}_i^\top. \end{aligned}$$

In the second step, a normalized vector $\mathbf{u}_2(0) \in \mathbb{R}^N$ is chosen and $\mathbf{u}_2(1), \mathbf{u}_2(2), \dots, \mathbf{u}_2(T_2)$ are obtained as follows:

$$\mathbf{u}_2(t+1) = \frac{S_2 \mathbf{u}_2(t)}{\|S_2 \mathbf{u}_2(t)\|}, \quad t = 0, 1, \dots, T_2 - 1.$$

The normalized vector $\mathbf{u}_2(T_2)$ is considered as an approximate eigenvector of S associated with λ_2 . Thus $\mathbf{u}_2(T_2)$ is set to the second column of U , and the second principal component is subtracted from S_2 . This process is repeated until P columns of U are obtained.

III. PROPOSED METHOD FOR PCA

A. Update Rule

Although the power method is simple and easy to implement, it has to obtain principal components one by one. Since the computation process for determining each principal component must be terminated after a finite number of iterations, the obtained value is slightly different from the true value. Furthermore, these errors propagate to computations of the subsequent principal components. To address this issue, we propose in this paper a novel iterative method for PCA that can obtain the specified number of principal components in parallel.

The method we propose in this paper is motivated by the continuous-time and discrete-time power methods [15], [16] for estimating the algebraic connectivity of a graph, and is described by

$$\begin{aligned} & \mathbf{u}_i(t+1) \\ &= \mathbf{u}_i(t) + \epsilon \left(-k_1 \sum_{j=1}^i \mathbf{u}_j(t) \mathbf{u}_j(t)^\top \mathbf{u}_i(t) + k_2 S \mathbf{u}_i(t) \right), \\ & \quad i = 1, 2, \dots, P \end{aligned} \quad (4)$$

where $\mathbf{u}_i(t) \in \mathbb{R}^N$ is the estimated value of the i -th principal component after t iterations, and k_1 and k_2 are positive constants. The term $-k_1 \sum_{j=1}^i \mathbf{u}_j(t) \mathbf{u}_j(t)^\top \mathbf{u}_i(t)$ is divided into

$$-k_1 \sum_{j=1}^{i-1} \mathbf{u}_j(t) \mathbf{u}_j(t)^\top \mathbf{u}_i(t)$$

and

$$-k_1 \mathbf{u}_i(t) \mathbf{u}_i(t)^\top \mathbf{u}_i(t) = -k_1 \|\mathbf{u}_i(t)\|^2 \mathbf{u}_i(t).$$

The former moves $\mathbf{u}_i(t)$ in a direction orthogonal to $\mathbf{u}_1(t), \mathbf{u}_2(t), \dots, \mathbf{u}_{i-1}(t)$, and the latter toward the origin. The term $k_2 S \mathbf{u}_i(t)$ moves $\mathbf{u}_i(t)$ toward an eigenvector of S associated with the largest eigenvalue λ_1 just like the power method.

In order to test the validity of the update rule (4), the authors implemented it in Python and examined the behavior of $\mathbf{u}_1(t), \mathbf{u}_2(t), \dots, \mathbf{u}_P(t)$ for various matrices X generated using uniform random numbers. Figure 1 shows a result for the case where $N = 10$, $L = 15$, $P = 6$, $k_1 = 0.2$, $k_2 = 0.4$ and $\epsilon = 0.1$. Time evolution of the absolute value of the cosine similarity of $\mathbf{u}_i(t)$ and $\mathbf{q}_i$, which is defined by $|\mathbf{q}_i^\top \mathbf{u}_i(t)| / \|\mathbf{u}_i(t)\|$, for $i = 1, 2, \dots, 6$ is shown in Fig. 1(a), and a zoomed view of Fig. 1(a) is shown in Fig. 1(b). It is

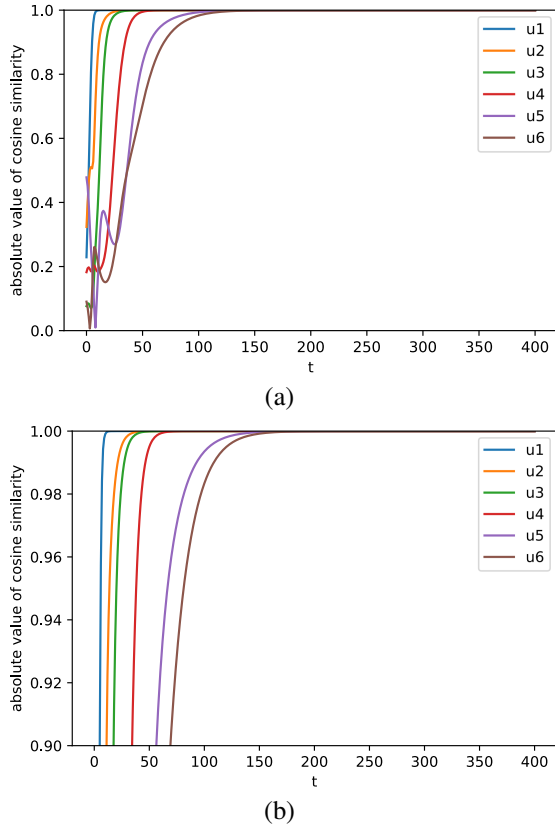


Fig. 1. Convergence of $\mathbf{u}_1(t), \mathbf{u}_2(t), \dots, \mathbf{u}_6(t)$. (a) Time evolution of the absolute value of the cosine similarity between $\mathbf{u}_i(t)$ and $\mathbf{q}_i$ for $i = 1, 2, \dots, 6$. (b) Zoomed view of (a).

seen from these figures that all of the six curves converge to one, which means that the proposed method can obtain the six principal components in parallel. It is also seen from the figures that $\mathbf{u}_1(t)$ converges first to the first principal component, followed by $\mathbf{u}_2(t), \mathbf{u}_3(t), \mathbf{u}_4(t), \mathbf{u}_5(t)$, and finally $\mathbf{u}_6(t)$ converging to the sixth principal component.

IV. THEORETICAL ANALYSIS

A. Equilibrium Point Analysis

In this section, we perform theoretical analysis to guarantee the validity of the proposed method (4) partially. The first step is to find all equilibrium points.

Theorem 1. A point $(\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_P) \in \mathbb{R}^{N \times P}$ is an equilibrium point of (4) if and only if the following two conditions are satisfied.

- 1) For all $i \in \{1, 2, \dots, P\}$, $\mathbf{u}_i$ is a member of the set

$$E := \left\{ \mathbf{0}, \pm \sqrt{\frac{k_2 \lambda_1}{k_1}} \mathbf{q}_1, \pm \sqrt{\frac{k_2 \lambda_2}{k_1}} \mathbf{q}_2, \dots, \pm \sqrt{\frac{k_2 \lambda_N}{k_1}} \mathbf{q}_N \right\}$$

where $\mathbf{q}_j$ is the j -th column of the orthogonal matrix $\mathbf{Q}$, which is one of the two normalized eigenvectors of $\mathbf{S}$ associated with the eigenvalue λ_j .

- 2) For each $j \in \{1, 2, \dots, N\}$, there exists at most one $i \in \{1, 2, \dots, P\}$ such that $\mathbf{u}_i \in \{\pm \sqrt{k_2 \lambda_j / k_1} \mathbf{q}_j\}$.

Proof. A point $(\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_P)$ is an equilibrium point of (4) if and only if

$$\mathbf{S} \mathbf{u}_i = \frac{k_1}{k_2} \sum_{j=1}^i \mathbf{u}_j \mathbf{u}_j^\top \mathbf{u}_i, \quad i = 1, 2, \dots, P. \quad (5)$$

From the first equation in (5), we have

$$\mathbf{S} \mathbf{u}_1 = \frac{k_1}{k_2} \mathbf{u}_1 \|\mathbf{u}_1\|^2. \quad (6)$$

Hence $\mathbf{u}_1$ is either $\mathbf{0}$ or an eigenvector of $\mathbf{S}$. Substituting $\mathbf{u}_1 = c \mathbf{q}_j$ with $c \neq 0$ and $j \in \{1, 2, \dots, N\}$ into (6), we have $c \lambda_j \mathbf{q}_j = (c^3 k_1 / k_2) \mathbf{q}_j$, which is satisfied if and only if $c = \pm \sqrt{k_2 \lambda_j / k_1}$. Therefore the set of all solutions to (6) is given by E .

From the second equation in (5), we have

$$\mathbf{S} \mathbf{u}_2 = \frac{k_1}{k_2} \mathbf{u}_1 \mathbf{u}_1^\top \mathbf{u}_2 + \frac{k_1}{k_2} \mathbf{u}_2 \|\mathbf{u}_2\|^2$$

which can be rewritten as

$$\left(\mathbf{S} - \frac{k_1}{k_2} \mathbf{u}_1 \mathbf{u}_1^\top \right) \mathbf{u}_2 = \frac{k_1}{k_2} \mathbf{u}_2 \|\mathbf{u}_2\|^2. \quad (7)$$

Hence $\mathbf{u}_2$ is either $\mathbf{0}$ or an eigenvector of $\mathbf{S} - (k_1/k_2) \mathbf{u}_1 \mathbf{u}_1^\top$. If $\mathbf{u}_1 = \mathbf{0}$, by the same discussion as above, we can say that the set of all solutions to (7) is given by E . If $\mathbf{u}_1 \in \{\pm \sqrt{k_2 \lambda_{j_1} / k_1} \mathbf{q}_{j_1}\}$ for some $j_1 \in \{1, 2, \dots, N\}$ then (7) reduces to

$$(\mathbf{S} - \lambda_{j_1} \mathbf{q}_{j_1} \mathbf{q}_{j_1}^\top) \mathbf{u}_2 = \frac{k_1}{k_2} \mathbf{u}_2 \|\mathbf{u}_2\|^2. \quad (8)$$

Note that $\mathbf{q}_1, \dots, \mathbf{q}_{j_1-1}, \mathbf{q}_{j_1+1}, \dots, \mathbf{q}_N$ are normalized eigenvectors of the matrix $\mathbf{S} - \lambda_{j_1} \mathbf{q}_{j_1} \mathbf{q}_{j_1}^\top$ associated with its eigenvalues $\lambda_1, \dots, \lambda_{j_1-1}, 0, \lambda_{j_1+1}, \dots, \lambda_N$, respectively. Note also that $\mathbf{u}_2 = c \mathbf{q}_{j_1}$ ($c \neq 0$) cannot be a solution to (8). Therefore the set of all solutions to (8) is given by $E \setminus \{\pm \sqrt{k_2 \lambda_{j_1} / k_1} \mathbf{q}_{j_1}\}$.

Continuing the above discussion for the remaining equations, we conclude that if $(\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_P)$ is an equilibrium point of (4) then the two conditions are satisfied. Conversely, if the two conditions are satisfied then it is clear that $(\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_P)$ is an equilibrium point of (4). $\square$

B. Stability of Equilibrium Points

The second step of our theoretical analysis is to determine local stability of each equilibrium point. Substituting (2) into (4) and multiplying both sides by $\mathbf{Q}^\top$ on the left, we have

$$\begin{aligned} \mathbf{Q}^\top \mathbf{u}_i(t+1) = \mathbf{Q}^\top \mathbf{u}_i(t) + \epsilon \left(-k_1 \sum_{j=1}^i \mathbf{Q}^\top \mathbf{u}_j(t) \mathbf{u}_j(t)^\top \mathbf{u}_i(t) \right. \\ \left. + k_2 \mathbf{\Lambda} \mathbf{Q}^\top \mathbf{u}_i(t) \right), \quad i = 1, 2, \dots, P. \end{aligned}$$

Changing the variable using $\mathbf{v}_i(t) := \mathbf{Q}^\top \mathbf{u}_i(t)$ for $i = 1, 2, \dots, P$, we obtain a new update rule described by

$$\begin{aligned} \mathbf{v}_i(t+1) = \mathbf{v}_i(t) + \epsilon \left(-k_1 \sum_{j=1}^i \mathbf{v}_j(t) \mathbf{v}_j(t)^\top \mathbf{v}_i(t) + k_2 \mathbf{\Lambda} \mathbf{v}_i(t) \right) \\ i = 1, 2, \dots, P. \quad (9) \end{aligned}$$

Let $(\mathbf{u}_1^*, \mathbf{u}_2^*, \dots, \mathbf{u}_P^*)$ be an equilibrium point of (4). Then $(\mathbf{v}_1^*, \mathbf{v}_2^*, \dots, \mathbf{v}_P^*)$ with $\mathbf{v}_i^* = \mathbf{Q}^\top \mathbf{u}_i^*$ for $i = 1, 2, \dots, P$ is an equilibrium point of (9), that is, $(\mathbf{v}_1^*, \mathbf{v}_2^*, \dots, \mathbf{v}_P^*)$ satisfies

$$-k_1 \sum_{j=1}^i \mathbf{v}_j^* \mathbf{v}_j^{*\top} \mathbf{v}_i^* + k_2 \mathbf{\Lambda} \mathbf{v}_i^* = \mathbf{0}, \quad i = 1, 2, \dots, P. \quad (10)$$

We now obtain the linearized update rule around this equilibrium point. Substituting $\mathbf{v}_i(t) = \mathbf{v}_i^* + \Delta \mathbf{v}_i(t)$ into (9) and ignoring the second and third order terms of $\Delta \mathbf{v}_1(t), \Delta \mathbf{v}_2(t), \dots, \Delta \mathbf{v}_P(t)$, we have

$$\begin{aligned} \Delta \mathbf{v}_i(t+1) &= \Delta \mathbf{v}_i(t) + \epsilon \left(-k_1 \sum_{j=1}^i (\mathbf{v}_j^* + \Delta \mathbf{v}_j(t)) (\mathbf{v}_j^* + \Delta \mathbf{v}_j(t))^\top \right. \\ &\quad \left. (\mathbf{v}_i^* + \Delta \mathbf{v}_i(t)) + k_2 \mathbf{\Lambda} (\mathbf{v}_i^* + \Delta \mathbf{v}_i(t)) \right) \\ &\approx \Delta \mathbf{v}_i(t) + \epsilon \left(-k_1 \sum_{j=1}^i (\Delta \mathbf{v}_j(t) \mathbf{v}_j^{*\top} \mathbf{v}_i^* \right. \\ &\quad \left. + \mathbf{v}_j^* \Delta \mathbf{v}_j(t)^\top \mathbf{v}_i^* + \mathbf{v}_j^* \mathbf{v}_j^{*\top} \Delta \mathbf{v}_i(t)) + k_2 \mathbf{\Lambda} \Delta \mathbf{v}_i(t) \right) \\ &= -\epsilon k_1 \sum_{j=1}^{i-1} \left(\mathbf{v}_j^{*\top} \mathbf{v}_i^* \mathbf{I} + \mathbf{v}_j^* \mathbf{v}_i^{*\top} \right) \Delta \mathbf{v}_j(t) \\ &\quad + \left(\mathbf{I} - \epsilon k_1 \left(\|\mathbf{v}_i^*\|^2 \mathbf{I} + \mathbf{v}_i^* \mathbf{v}_i^{*\top} + \sum_{j=1}^i \mathbf{v}_j^* \mathbf{v}_j^{*\top} \right) \right. \\ &\quad \left. + \epsilon k_2 \mathbf{\Lambda} \right) \Delta \mathbf{v}_i(t) \end{aligned} \quad (11)$$

where we have used (10).

Theorem 2. *If the positive constant ϵ satisfies*

$$\epsilon < \frac{1}{k_2 \lambda_1} \quad (12)$$

then only those equilibrium points $(\mathbf{u}_1^, \mathbf{u}_2^*, \dots, \mathbf{u}_P^*)$ of (4) such that $\mathbf{u}_i^* \in \{\pm \sqrt{k_2 \lambda_i / k_1} \mathbf{q}_i\}$ for $i = 1, 2, \dots, P$ are stable and all other equilibrium points are unstable.*

Proof. An equilibrium point $(\mathbf{u}_1^*, \mathbf{u}_2^*, \dots, \mathbf{u}_P^*)$ of (4) is stable if and only if the corresponding equilibrium point $(\mathbf{v}_1^*, \mathbf{v}_2^*, \dots, \mathbf{v}_P^*)$ of (9), which is given by $\mathbf{v}_i^* = \mathbf{Q}^\top \mathbf{u}_i^*$ for $i = 1, 2, \dots, P$, is stable.

We first consider an equilibrium point $(\mathbf{u}_1^*, \mathbf{u}_2^*, \dots, \mathbf{u}_P^*)$ of (4) such that $\mathbf{u}_i^* \in \{\pm \sqrt{k_2 \lambda_i / k_1} \mathbf{q}_i\}$ for $i = 1, 2, \dots, P$. The corresponding equilibrium point of (9) is given by $\mathbf{v}_i^* = \mathbf{Q}^\top \mathbf{u}_i^* \in \{\pm \sqrt{k_2 \lambda_i / k_1} \mathbf{e}_i\}$ for $i = 1, 2, \dots, P$, where $\mathbf{e}_i$ is the vector of which the i -th entry is 1 and other entries are 0. Substituting $\mathbf{v}_i^* = \sqrt{k_2 \lambda_i / k_1} \mathbf{e}_i$ or $\mathbf{v}_i^* = -\sqrt{k_2 \lambda_i / k_1} \mathbf{e}_i$ for

$i = 1, 2, \dots, P$ into (11), we have

$$\begin{aligned} &\begin{pmatrix} \Delta \mathbf{v}_1(t+1) \\ \Delta \mathbf{v}_2(t+1) \\ \vdots \\ \Delta \mathbf{v}_P(t+1) \end{pmatrix} \\ &= \begin{pmatrix} \mathbf{A}_{11} & \mathbf{O} & \cdots & \mathbf{O} \\ \mathbf{A}_{21} & \mathbf{A}_{22} & \ddots & \vdots \\ \vdots & \vdots & \ddots & \mathbf{O} \\ \mathbf{A}_{P1} & \mathbf{A}_{P2} & \cdots & \mathbf{A}_{PP} \end{pmatrix} \begin{pmatrix} \Delta \mathbf{v}_1(t) \\ \Delta \mathbf{v}_2(t) \\ \vdots \\ \Delta \mathbf{v}_P(t) \end{pmatrix} \end{aligned} \quad (13)$$

where

$$\begin{aligned} \mathbf{A}_{ii} &= \mathbf{I} - \epsilon k_1 \left(\|\mathbf{v}_i^*\|^2 \mathbf{I} + \mathbf{v}_i^* \mathbf{v}_i^{*\top} + \sum_{j=1}^i \mathbf{v}_j^* \mathbf{v}_j^{*\top} \right) + \epsilon k_2 \mathbf{\Lambda} \\ &= \mathbf{I} - \epsilon k_1 \left(\frac{k_2 \lambda_i}{k_1} \mathbf{I} + \frac{k_2 \lambda_i}{k_1} \mathbf{e}_i \mathbf{e}_i^\top + \sum_{j=1}^i \frac{k_2 \lambda_j}{k_1} \mathbf{e}_j \mathbf{e}_j^\top \right) \\ &\quad + \epsilon k_2 \mathbf{\Lambda} \\ &= \mathbf{I} - \epsilon k_2 \left(\lambda_i \mathbf{I} + \lambda_i \mathbf{e}_i \mathbf{e}_i^\top + \sum_{j=1}^i \lambda_j \mathbf{e}_j \mathbf{e}_j^\top - \mathbf{\Lambda} \right) \end{aligned}$$

for $i = 1, 2, \dots, P$. Note that $\mathbf{A}_{ii}$ is a diagonal matrix of which the j -th diagonal entry is given by $1 - \epsilon k_2 \lambda_i$ if $1 \leq j \leq i-1$, $1 - 2\epsilon k_2 \lambda_i$ if $j = i$, and $1 - \epsilon k_2 (\lambda_i - \lambda_j)$ if $i+1 \leq j \leq N$. We see from this fact that the coefficient matrix on the right-hand side of (13) is lower triangular, and has diagonal entries with absolute values less than 1 if $\epsilon k_2 \lambda_1 < 1$. Therefore, the equilibrium point is stable if $\epsilon k_2 \lambda_1 < 1$.

We next consider an equilibrium point $(\mathbf{u}_1^*, \mathbf{u}_2^*, \dots, \mathbf{u}_P^*)$ of (4) such that $\mathbf{u}_i^* \in \{\pm \sqrt{k_2 \lambda_i / k_1} \mathbf{q}_i\}$ for $i = 1, 2, \dots, r-1$ and $\mathbf{u}_r^* \notin \{\pm \sqrt{k_2 \lambda_r / k_1} \mathbf{q}_r\}$, where $1 \leq r \leq P$. It follows from Theorem 1 that

$$\mathbf{u}_r^* \in \left\{ \mathbf{0}, \pm \sqrt{\frac{k_2 \lambda_{r+1}}{k_1}} \mathbf{q}_{r+1}, \dots, \pm \sqrt{\frac{k_2 \lambda_N}{k_1}} \mathbf{q}_N \right\}.$$

The corresponding equilibrium point of (9) satisfies $\mathbf{v}_i^* \in \{\pm \sqrt{k_2 \lambda_i / k_1} \mathbf{e}_i\}$ for $i = 1, 2, \dots, r-1$ and

$$\mathbf{v}_r^* \in \left\{ \mathbf{0}, \pm \sqrt{\frac{k_2 \lambda_{r+1}}{k_1}} \mathbf{e}_{r+1}, \dots, \pm \sqrt{\frac{k_2 \lambda_N}{k_1}} \mathbf{e}_N \right\}.$$

Hence the r -th diagonal entry of $\mathbf{A}_{rr}$ in (13), which is denoted as $(\mathbf{A}_{rr})_{rr}$, satisfies

$$\begin{aligned} (\mathbf{A}_{rr})_{rr} &= 1 - \epsilon k_1 \left(\|\mathbf{v}_r^*\|^2 + 2 \left(\mathbf{v}_r^* \mathbf{v}_r^{*\top} \right)_{rr} \right) + \epsilon k_2 \lambda_r \\ &= 1 - \epsilon k_1 \|\mathbf{v}_r^*\|^2 + \epsilon k_2 \lambda_r \\ &\geq 1 - \epsilon k_1 \cdot \frac{k_2 \lambda_{r+1}}{k_1} + \epsilon k_2 \lambda_r \\ &= 1 + \epsilon k_2 (\lambda_r - \lambda_{r+1}) \\ &> 1. \end{aligned}$$

This means that the equilibrium point is unstable. $\square$

C. Boundedness of Solutions

The third step of our theoretical analysis is to guarantee that any sequence $\{(\mathbf{u}_1(t), \mathbf{u}_2(t), \dots, \mathbf{u}_P(t))\}_{t=0}^\infty$ generated by the proposed method (4) is bounded.

Theorem 3. *Let $\{(\mathbf{u}_1(t), \mathbf{u}_2(t), \dots, \mathbf{u}_P(t))\}_{t=0}^\infty$ be a sequence generated by the update rule (4). If the positive constant ϵ satisfies*

$$\epsilon \leq \frac{1}{2Pk_2\lambda_1} \quad (14)$$

and the inequalities

$$\|\mathbf{u}_i(t)\|^2 \leq \frac{2 + \epsilon k_2 \lambda_N}{P\epsilon k_1}, \quad i = 1, 2, \dots, P \quad (15)$$

are satisfied at $t = 0$ then (15) is satisfied for all $t \geq 0$.

Proof. We first show that if ϵ satisfies (14) and $\mathbf{u}_1(t), \mathbf{u}_2(t), \dots, \mathbf{u}_P(t)$ satisfy (15) then $\|\mathbf{u}_i(t+1)\|^2 \leq \|\mathbf{u}_i(t)\|^2$ for any $i \in \{1, 2, \dots, P\}$ such that

$$\|\mathbf{u}_i(t)\|^2 \geq \frac{k_2 \lambda_1}{k_1}. \quad (16)$$

Note that, under the assumption (14), the right-hand side of (16) is less than that of (15) because

$$\begin{aligned} \frac{2 + \epsilon k_2 \lambda_N}{P\epsilon k_1} - \frac{k_2 \lambda_1}{k_1} &= \frac{1}{P\epsilon k_1} (2 - \epsilon k_2 (P\lambda_1 - \lambda_N)) \\ &\geq \frac{1}{P\epsilon k_1} \left(2 - \frac{1}{2Pk_2\lambda_1} \cdot k_2 (P\lambda_1 - \lambda_N) \right) \\ &\geq \frac{1}{P\epsilon k_1} \left(\frac{3}{2} + \frac{\lambda_N}{2P\lambda_1} \right) \\ &> 0. \end{aligned}$$

Changing the variable using $\mathbf{v}_i(t) := \mathbf{Q}^\top \mathbf{u}_i(t)$, and taking (9) into account, we have

$$\mathbf{v}_i(t+1) = (\mathbf{I} + \epsilon \mathbf{B}_i(t)) \mathbf{v}_i(t)$$

where

$$\mathbf{B}_i(t) := -k_1 \sum_{j=1}^{i-1} \mathbf{v}_j(t) \mathbf{v}_j(t)^\top - k_1 \|\mathbf{v}_i(t)\|^2 \mathbf{I} + k_2 \mathbf{\Lambda}.$$

Since $\mathbf{B}_i(t)$ is a symmetric matrix, there exist an orthogonal matrix $\mathbf{C}_i(t) \in \mathbb{R}^{N \times N}$ and a diagonal matrix $\mathbf{D}_i(t)$ such that $\mathbf{B}_i(t) = \mathbf{C}_i(t) \mathbf{D}_i(t) \mathbf{C}_i(t)^\top$, where the diagonal entries of $\mathbf{D}_i(t)$ are the eigenvalues of $\mathbf{B}_i(t)$. Using this matrix decomposition, we have

$$\begin{aligned} \|\mathbf{v}_i(t+1)\|^2 - \|\mathbf{v}_i(t)\|^2 &= \mathbf{v}_i(t)^\top (\mathbf{I} + \epsilon \mathbf{B}_i(t))^2 \mathbf{v}_i(t) - \mathbf{v}_i(t)^\top \mathbf{v}_i(t) \\ &= \epsilon \mathbf{v}_i(t)^\top (2\mathbf{I} + \epsilon \mathbf{B}_i(t)) \mathbf{B}_i(t) \mathbf{v}_i(t) \\ &= \epsilon \mathbf{v}_i(t)^\top (2\mathbf{I} + \epsilon \mathbf{C}_i(t) \mathbf{D}_i(t) \mathbf{C}_i(t)^\top) \\ &\quad (\mathbf{C}_i(t) \mathbf{D}_i(t) \mathbf{C}_i(t)^\top) \mathbf{v}_i(t) \\ &= \epsilon \mathbf{v}_i(t)^\top \mathbf{C}_i(t) (2\mathbf{I} + \epsilon \mathbf{D}_i(t)) \mathbf{D}_i(t) \mathbf{C}_i(t)^\top \mathbf{v}_i(t) \\ &= \epsilon (\mathbf{C}_i(t)^\top \mathbf{v}_i(t))^\top (2\mathbf{I} + \epsilon \mathbf{D}_i(t)) \\ &\quad \mathbf{D}_i(t) (\mathbf{C}_i(t)^\top \mathbf{v}_i(t)). \end{aligned} \quad (17)$$

Here, under the assumption (16), an upper bound for the diagonal entries of $\mathbf{D}_i(t)$ is given by

$$\begin{aligned} \max_{\|\mathbf{x}\|=1} \mathbf{x}^\top \mathbf{B}_i(t) \mathbf{x} &= \max_{\|\mathbf{x}\|=1} \mathbf{x}^\top \left(-k_1 \sum_{j=1}^{i-1} \mathbf{v}_j(t) \mathbf{v}_j(t)^\top \right. \\ &\quad \left. - k_1 \|\mathbf{v}_i(t)\|^2 \mathbf{I} + k_2 \mathbf{\Lambda} \right) \mathbf{x} \\ &= \max_{\|\mathbf{x}\|=1} \left(-k_1 \sum_{j=1}^{i-1} (\mathbf{x}^\top \mathbf{v}_j(t))^2 \right. \\ &\quad \left. - k_1 \|\mathbf{v}_i(t)\|^2 \|\mathbf{x}\|^2 + k_2 \mathbf{x}^\top \mathbf{\Lambda} \mathbf{x} \right) \\ &\leq -k_1 \|\mathbf{v}_i(t)\|^2 + k_2 \lambda_1 \\ &\leq -k_1 \cdot \frac{k_2 \lambda_1}{k_1} + k_2 \lambda_1 \\ &= 0. \end{aligned} \quad (18)$$

Similarly, under the assumption (15), a lower bound for the diagonal entries of $\mathbf{D}_i(t)$ is given by

$$\begin{aligned} \min_{\|\mathbf{x}\|=1} \mathbf{x}^\top \mathbf{B}_i(t) \mathbf{x} &= \min_{\|\mathbf{x}\|=1} \left(-k_1 \sum_{j=1}^{i-1} (\mathbf{x}^\top \mathbf{v}_j(t))^2 \right. \\ &\quad \left. - k_1 \|\mathbf{v}_i(t)\|^2 \|\mathbf{x}\|^2 + k_2 \mathbf{x}^\top \mathbf{\Lambda} \mathbf{x} \right) \\ &\geq -k_1 \sum_{j=1}^{i-1} \|\mathbf{v}_j(t)\|^2 - k_1 \|\mathbf{v}_i(t)\|^2 + k_2 \lambda_N \\ &\geq -k_1 i \cdot \frac{2 + \epsilon k_2 \lambda_N}{P\epsilon k_1} + k_2 \lambda_N \\ &\geq -\frac{2 + \epsilon k_2 \lambda_N}{\epsilon} + k_2 \lambda_N \\ &= -\frac{2}{\epsilon} \end{aligned} \quad (19)$$

where we have used the Cauchy-Schwarz inequality $(\mathbf{x}^\top \mathbf{v}_j(t))^2 \leq \|\mathbf{x}\|^2 \|\mathbf{v}_j(t)\|^2$. We see from (18) and (19) that the diagonal entries of $\mathbf{D}_i(t)$ are always in the closed interval $[-2/\epsilon, 0]$, which implies that the diagonal entries of the diagonal matrix $(2\mathbf{I} + \epsilon \mathbf{D}_i(t)) \mathbf{D}_i(t)$ are nonpositive. Therefore it follows from (17) that $\|\mathbf{v}_i(t+1)\|^2 \leq \|\mathbf{v}_i(t)\|^2$, which means that $\|\mathbf{u}_i(t+1)\|^2 \leq \|\mathbf{u}_i(t)\|^2$.

We next show that if ϵ satisfies (14) and $\mathbf{u}_1(t), \mathbf{u}_2(t), \dots, \mathbf{u}_P(t)$ satisfy (15) then

$$\|\mathbf{u}_i(t+1)\|^2 < \frac{2 + \epsilon k_2 \lambda_N}{P\epsilon k_1} \quad (20)$$

holds for any $i \in \{1, 2, \dots, P\}$ such that

$$\|\mathbf{u}_i(t)\|^2 < \frac{k_2 \lambda_1}{k_1}. \quad (21)$$

As in the first part, we change the variable using $\mathbf{v}_i(t) := \mathbf{Q}^\top \mathbf{u}_i(t)$. Under the assumption (21), an upper bound for the

diagonal entries of $D_i(t)$ is given by $k_2\lambda_1$, and a lower bound is given by

$$\begin{aligned}
& \min_{\|x\|=1} x^\top B_i(t) x \\
& \geq -k_1 \sum_{j=1}^{i-1} \|v_j(t)\|^2 - k_1 \|v_i(t)\|^2 + k_2\lambda_N \\
& > -k_1(i-1) \cdot \frac{2 + \epsilon k_2\lambda_N}{P\epsilon k_1} - k_1 \cdot \frac{k_2\lambda_1}{k_1} + k_2\lambda_N \\
& \geq -\frac{P-1}{P} \left(\frac{2}{\epsilon} - k_2\lambda_N \right) - k_2(\lambda_1 - \lambda_N) \\
& \geq -\frac{2}{\epsilon} + \frac{2}{P\epsilon} - k_2(\lambda_1 - \lambda_N) \\
& \geq -\frac{2}{\epsilon} + \frac{4Pk_2\lambda_1}{P} - k_2(\lambda_1 - \lambda_N) \\
& = -\frac{2}{\epsilon} + k_2(3\lambda_1 + \lambda_N) \\
& > -\frac{2}{\epsilon}
\end{aligned}$$

where we have used (14). We thus have an upper bound for $\|v_i(t+1)\|^2$ as follows:

$$\begin{aligned}
\|v_i(t+1)\|^2 &= \|v_i(t)\|^2 + \epsilon (C_i(t)^\top v_i(t))^\top (2I + \epsilon D_i(t)) \\
&\quad D_i(t) (C_i(t)^\top v_i(t)) \\
&\leq \|v_i(t)\|^2 + \epsilon(2 + \epsilon k_2\lambda_1) k_2\lambda_1 \|v_i(t)\|^2 \\
&= (1 + \epsilon k_2\lambda_1)^2 \|v_i(t)\|^2 \\
&< (1 + \epsilon k_2\lambda_1)^2 \cdot \frac{k_2\lambda_1}{k_1}.
\end{aligned}$$

This upper bound is less than the right-hand side of (20) because

$$\begin{aligned}
& \frac{2 + \epsilon k_2\lambda_N}{P\epsilon k_1} - (1 + \epsilon k_2\lambda_1)^2 \cdot \frac{k_2\lambda_1}{k_1} \\
&= \frac{1}{P\epsilon k_1} (2 + \epsilon k_2\lambda_N - P\epsilon k_2\lambda_1(1 + \epsilon k_2\lambda_1)^2) \\
&\geq \frac{1}{P\epsilon k_1} (2 - P\epsilon k_2\lambda_1(1 + \epsilon k_2\lambda_1)^2) \\
&\geq \frac{1}{P\epsilon k_1} \left(2 - \frac{1}{2} \left(1 + \frac{1}{2P} \right)^2 \right) \\
&\geq \frac{1}{P\epsilon k_1} \left(2 - \frac{1}{2} \left(1 + \frac{1}{2} \right)^2 \right) \\
&= \frac{7}{8P\epsilon k_1} \\
&> 0
\end{aligned}$$

where we have used (14) and $P \geq 1$. Therefore $\|v_i(t+1)\|^2$ is less than the right-hand side of (20). $\square$

V. CONCLUSIONS

A novel iterative method for finding principal components from a given data matrix was proposed in this paper. The proposed method is based on the continuous-time power method for computing the algebraic connectivity of a graph in a distributed manner. Unlike traditional PCA algorithms such

as the power method, which computes principal components one by one, the proposed method can obtain multiple principal components in parallel. The validity of the proposed method was shown through numerical experiments. In addition, theoretical analysis was performed to find equilibrium points, determine their local stability, and show the boundedness of solutions. One of the future challenges is to prove the convergence of solutions.

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